

Red vs Green in Motion Perception — Literature Summary

*VRDots context: red dots and green dots on a black background.
Neither color is isoluminant relative to the background.
Question: does color (red vs green) systematically affect motion detection?*

1. Motion perception is luminance-contrast driven, not color-specific

The magnocellular (M-pathway) stream dominates first-order motion processing. M-cells are broadband — they sum L and M cone signals nearly equally and respond primarily to luminance contrast, not chromatic contrast. Because red (#CC3333) and green (#228B22) dots both have substantial luminance contrast against a black background, both stimulate the M-pathway effectively. The canonical 'motion is blind to color' finding (Cavanagh, Tyler & Favreau 1984; Ramachandran & Gregory 1978; Livingstone & Hubel 1987) refers specifically to isoluminant stimuli — those with zero luminance contrast against their background. This condition does not apply in VRDots.

When luminance contrast is present, chromatic identity (red vs green) is largely irrelevant to the motion energy computation itself. The motion signal is carried by the luminance increment of each dot against the black surround. Both red and green dots, if matched in physical luminance, should drive motion perception equivalently through the M-pathway.

2. Luminance asymmetry: red and green are not equal on a display

Even at equal RGB digital values, red and green stimuli differ in perceived luminance. The photopic luminosity function $V(\lambda)$ peaks near 555 nm (yellow-green). Green LEDs (peak ~530 nm) fall near the photopic peak; red LEDs (peak ~630–640 nm) fall on the descending limb and have lower luminous efficiency. Consequently, a red patch and a green patch at equal RGB values will typically differ by 30–60% in perceived luminance, with green appearing brighter.

Display gamma correction partially compensates but is calibrated for overall white balance, not individual channel perceptual equality. On the Meta Quest specifically, the display is a transmissive LCD (Quest 2) or pancake-lens LCD (Quest 3) with RGB sub-pixel layout; the green channel typically contributes ~59% of the luminous flux for a white pixel. If the VRDots stimulus uses equal RGB values for red and green dots, the green dots are likely brighter in photopic units. This could confer a modest motion detection advantage on green, not red.

Practical implication: if a systematic color advantage exists in the VRDots data, it is more likely to favor green than red, due to its proximity to the peak of the photopic luminosity function. Any red advantage would need an attentional rather than a sensory-contrast explanation.

Red vs Green in Motion Perception — p.2: Attention & Segmentation

3. Red attention salience — documented but attention-stage, not motion-stage

Red stimuli reliably capture attention faster and more strongly than other colors in visual search and cueing paradigms, even when luminance is controlled. This effect is robust across cultures and has been linked to evolutionary pressure (ripe fruit, blood, predator coloration).

Key findings:

- Theeuwes & Kooi (1994): color singletons capture attention; red captures faster than green in free-field search.
- Anderson, Laurent & Yantis (2011, Psychological Science): reward-associated colors (typically trained with red) produce persistent attentional capture even when task-irrelevant.
- Elliot & Maier (2012, Current Directions in Psychological Science): red influences motivated behavior and attention across achievement and avoidance contexts.

Critically, this red capture effect operates at the attentional/post-selection stage, not at the level of the motion energy computation. In a task like VRDots where the observer must select one of two overlapping surfaces, red salience could bias initial attention capture toward the red surface — but this would be a Field-identity effect, not a motion-direction effect per se.

In VRDots, because red and green are balanced across conditions (which field is red vs green varies across trials), red salience would add noise but not a systematic directional bias at the group level. Within a trial, the observer might be more likely to initially attend to the red field — relevant if one is trying to interpret session-level variability or early vs late trial effects.

4. Color as a surface-segmentation cue for overlapping dot fields

The most directly relevant body of literature for VRDots is the role of color in segmenting overlapping transparent-motion surfaces. When two surfaces share a color, depth-plane filtering is weakened; when they differ in color, they are more easily segregated as distinct objects.

- Theeuwes, Atchley & Kramer (1998, Psychological Science): color differences between overlapping motion surfaces facilitate surface-based selection. Same-color surfaces are harder to segment. This is the primary motivation for R/G designs in VRDots.
- Croner & Albright (1999, Journal of Neuroscience): MT neurons in macaque showed color-selective surround suppression — suggesting color aids figure-ground segmentation even in motion-processing areas.
- Nakayama, He & Shimojo (1995): color is a preattentive grouping dimension comparable to depth and motion; surfaces that share a color are parsed as a single object more readily than surfaces differing in color.

These findings support using R/G in VRDots to strengthen surface segmentation, but do not predict a differential red vs green motion advantage. The benefit is symmetric: either color helps identify its surface; neither color inherently yields better motion detection than the other via the segmentation route.

Red vs Green in Motion Perception — p.3: Direct Comparisons & VRDots Implications

5. Direct red vs green motion detection comparisons

There is limited literature directly comparing red vs green in motion detection when both have luminance contrast. Most chromatic motion research compares isoluminant chromatic vs luminance stimuli, not red vs green at suprathreshold contrast.

- Gegenfurtner & Hawken (1996, Trends in Neurosciences): comprehensive review of color and motion; conclude that chromatic signals make a modest contribution to perceived speed and direction when combined with luminance signals, but the direction signal is dominated by luminance. No differential red/green advantage identified.
- Cavanagh & Anstis (1991, Vision Research): contribution of color to motion in normal and color-deficient observers. Motion from equiluminant color is weak but present; direction discrimination thresholds are elevated. Study does not isolate red vs green specifically.
- Lu & Sperling (2001, Journal of Vision): three-systems framework for motion perception — first-order (luminance), second-order (contrast modulation), third-order (salience/feature maps). Red salience could boost third-order motion tracking but this has not been directly tested for red vs green dot stimuli on black backgrounds.

Bottom line: no study known to this agent demonstrates a consistent red > green advantage in detecting or discriminating the direction of suprathreshold motion (i.e., where both colors have luminance contrast). The attentional red-capture literature is real but operates pre-translation; it does not predict differential heading discrimination.

6. Implications for VRDots

Given the above, the R/G design in VRDots is well-motivated (surface segmentation) and unlikely to introduce a systematic red-vs-green motion detection bias, provided:

- The two colors are approximately luminance-balanced on the Quest display.
- Red and green are counterbalanced across conditions (which field is red vs green varies across trials), neutralizing any residual color-salience bias at the group level.

Literature agent, VRDots — 2026-04-04. Citations reflect agent knowledge; verify before submission.

The most actionable concern is display luminance calibration. If green dots are substantially brighter than red dots in photopic units (as would follow from $V(\lambda)$), this could produce a systematic performance advantage for whichever surface is green. A flicker photometry or heterochromatic brightness matching check would resolve this. In the absence of calibration data, the R/G balance across conditions provides reasonable protection against this confound at the group level but not within a trial.

In the DepthColorLinked experiment specifically, the concern is whether Near=Red / Far=Green (or vice versa) introduces a color-salience confound with the depth manipulation. Per the experiment_status.md, color and depth are balanced across trials, so this is not a systematic confound — but it adds within-trial variance on trials where the salient-color surface is the non-target.

References (abbreviated)

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